Prüfbericht-Nr.: *Test report no.:*	**CN267M2L 001**	**Auftrags-Nr.:** *Order no.:*	**326194227**	Seite 1 von 29 *Page 1 of 29*
Kunden-Referenz-Nr.: *Client reference no.:*	**2097732**	**Auftragsdatum:** *Order date:*	2026-06-26	
Auftraggeber: *Client:*	REPT BATTERO Energy Co., Ltd. No.205, Binhai 6th Road, Wenzhouwan New District,Longwan District, Wenzhou 325000 Zhejiang, P.R.China			
Prüfgegenstand: *Test item:*	Rechargeable Li-ion Battery System			
Bezeichnung / Typ-Nr.: *Identification / Type no.:*	Powtrix2.2-0.5P-Y104R04C12-A, Powtrix2.2-0.5P-Y104R04C12-B, Powtrix2.2-0.5P-Y104R04C12-C, Powtrix2.2-0.5P-Y104R04C12-D, Powtrix2.2-0.25P-Y104R04C12-A, Powtrix2.2-0.25P-Y104R04C12-B, Powtrix2.2-0.25P-Y104R04C12-C, Powtrix2.2-0.25P-Y104R04C12-D			
Auftrags-Inhalt: *Order content:*	Assessment report			
Prüfgrundlage: *Test specification:*	This report provides evaluation and CFD analysis of the deflagration venting system in energy storage system referring to NFPA68:2023.			
Wareneingangsdatum: *Date of sample receipt:*	2025-07-08			
Prüfmuster-Nr.: *Test sample no:*	Engineering sample			
Prüfzeitraum: *Testing period:*	2025-07-08- 2025-07-25			
Ort der Prüfung: *Place of testing:*	TÜV Rheinland (Shanghai) Co., Ltd.			
Prüflaboratorium: *Testing laboratory:*	TÜV Rheinland (Shanghai) Co., Ltd.			
Prüfergebnis*: *Test result*:*	See main report			

geprüft von: *tested by:*	Rita Ren.	**genehmigt von:** *authorized by:*	Yichen
Datum: *Date:* 2026.08.19	Rita Ren	**Ausstellungsdatum:** *Issue date:* 2026.08.19	Yichen Zhao
Stellung */ Position:*	**Project Engineer**	**Stellung** */ Position:*	**Authorizer**

Sonstiges */ Other:*

Zustand des Prüfgegenstandes bei Anlieferung: *Condition of the test item at delivery:* Prüfmuster vollständig und unbeschädigt *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens.
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V05

1. Introduction:

TÜV Rheinland has been engaged by REPT BATTERO Energy Co., Ltd. to conduct an evaluation of deflagration venting performance for the Battery Energy Storage System (BESS) enclosure, which incorporates REPT BATTERO Energy Co., Ltd.'s Lithium Iron Phosphate model CB75 314Ah prismatic battery cells. The primary objective of this assessment, conducted in accordance with NFPA 68 guidelines, aimed at mitigating this risk through deflagration venting. It should be noted that site-specific details were not addressed in this design.

TÜV Rheinland utilized FLame ACceleration Simulator (FLACS) computational fluid dynamics (CFD) modeling to simulate the venting event resulting from the thermal runaway of the battery cells. This analysis also included an evaluation of the primary explosion hazard identified for this BESS, which arises from the release of flammable gases due to venting of Lithium iron phosphate (LFP) batteries. This report outlines a conceptual explosion protection design, developed in accordance with NFPA 855 and NFPA 68, aimed at mitigating this risk through deflagration venting.

The calculations pertaining to the deflagration venting in this analysis were based on the characteristics of the gas pressure The details were derived from the UL 9540A cell, module, and unit level testing, in conjunction with the NFPA 69 report, both of which were provided by REPT. These calculations were subsequently incorporated into the FLACS model to determine the gas flow patterns during venting within the enclosure.

The main performance objective of the analysis in this report was:

To follow the requirements of Section 6.3.1.1 of NFPA 68 2023 edition and assess performance criteria state that " P_{red} shall not exceed two-thirds of the ultimate strength for the vented enclosure, provided deformation of the equipment can be tolerated.".

In accordance with these objectives outlined in NFPA 68, vents must be designed to ensure that the maximum pressure generated within the enclosure during a vented deflagration remains below the enclosure strength by an adequate margin of safety to prevent structural failure.

2. Product Description:

2.1. General Description

The EUT covered by this report is one LFP lithium ion battery energy storage system. The system is available in two completely symmetrical arrangements, each of which contains one electrical cabinet, twelve battery racks and six switchgears. The electrical cabinet supports 4 forms of output options, which are 1-way output, 2-way output, 3-way output, 6-way output and 12-way output respectively. Each switchgear controls 2 racks respectively. The model of the rack is: Y104R04-0.5P, Y104R04-0.25P.

Each battery rack (Unit) has four battery modules. Each battery module contains 104 battery cells connected in series. Each battery module contains two BMU boards, which are used to measure and collect battery parameters and upload battery voltage and temperature information to the SBCU in the Switch Gear.

The EUT contains auxiliary distribution system, fire suppression system and chilling system.

The EUT is outdoor type. The insulation between the DC circuit and the metal enclosure is basic insulation. And the insulation between the DC circuit and communication ports is reinforced insulation or double insulation. OVC is considered for rack, it shall be isolated from an OVC III supply source(such as from an OVC III PCS) through an isolated transformer or protected in a manner that prevents transient over voltage conditions in end use.

The EUT supports multiple forms of output, and 1/2/3/6/12 output options can be selected to suit different PCS solutions.

The BMS functional safety was evaluated according to IEC 60730-1 Annex H and UL 60730-1 Annex H by TÜV Rheinland.

2.2. General Production Information:

2.2.1. Cell

The product information and parameters are provided by the client as below.

Product	Rechargeable Li-ion Battery Cell
Type/model	CB75
Capacity [Ah]	314
Cell Quantity	1
Nominal voltage [V]	3.2
Rated capacity [Wh]	1004.8
Upper limit charging voltage [V]	3.75
Maximum charging current [A]	314
Recommend charging current [A]	157
Maximum discharging current [A]	314
Recommend discharging current [A]	157
Continuous Maximum Power [W]	0.5P (502.4W)
Discharge cut-off voltage [V]	2.5（T＞0℃）；2.0（T≤0℃）
Temperature range for charging [°C]	0 to 60
Temperature range for discharging [°C]	-20 to 60
Overcharge protected voltage supply by battery system	3.75
Recommend charging method by manufacturer	Charge at constant current 314A until the voltage reaches 3.65V, then switch to constant voltage 3.65V till charge current drops to 16A
Dimension [mm]	Thickness (71.7±0.8) * Width (174.0±0.8) * Shoulder Height (204.4±0.8) mm * Total Height (206.8±0.8) mm
Weight [kg]	5.67±0.2kg
Cooling type	Liquid cooling

2.2.2.Module

The product information and parameters are provided by the client as below.

Product	Rechargeable Li-ion Battery Module	Rechargeable Li-ion Battery Module
Type/model	Y104-0.5P	Y104-0.25P
Capacity [Ah]	314	314
Cell Quantity	104	104
Battery structure	104S	104S
Nominal voltage [V]	332.8	332.8
Rated capacity [Wh]	104499.2	104499.2
Upper limit charging voltage [V]	379.6	379.6
Maximum charging current [A]	188	94.2
Recommend charging current [A]	157	78.5
Maximum discharging current [A]	188	94.2
Recommend discharging current [A]	157	78.5
Continuous Maximum Power [W]	0.5P (52249.6W)	0.25P (26124.8W)
Discharge cut-off voltage [V]	261.04	261.04
Temperature range for charging [°C]	1 to 55	1 to 55
Temperature range for discharging [°C]	-20 to 55	-20 to 55
Temperature threshold for protection	56	56
Overcharge protected voltage supply by battery system	⩾3.65V /Cell	⩾3.65V /Cell
Recommend charging method by manufacturer	After charging at a constant power of 0.5P until the maximum single-cell voltage reaches 3.55V, then charging at a constant power of 0.1P until the maximum single-cell voltage reaches 3.65V.	After charging at a constant power of 0.25P until the maximum single-cell voltage reaches 3.55V, then charging at a constant power of 0.1P until the maximum single-cell voltage reaches 3.65V.
Dimension [mm]	2150.5±5mm×779±2mm×250±2mm	2150.5±5mm×779±2mm×250±2mm
Weight [kg]	680kg±10kg	680kg±10kg
Ingress Protection (IP)	IP67	IP67
Cooling type	Liquid cooling	Liquid cooling
Altitude	3000m	3000m

2.2.3.Energy Storage System

The product information and parameters are provided by the client as below.

Product	**Rechargeable Li-ion Battery System**
Type/model	Powtrix2.2-0.5P-Y104R04C12-A, Powtrix2.2-0.5P-Y104R04C12-B, Powtrix2.2-0.5P-Y104R04C12-C, Powtrix2.2-0.5P-Y104R04C12-D
Capacity [Ah]	314*12
Cell Quantity	4992
Battery structure	416S12P
Nominal voltage [V]	1331.2
Rated capacity [Wh]	5015961.6
Upper limit charging voltage [V]	1500
Maximum charging current [A]	188*12
Recommend charging current [A]	157*12
Maximum discharging current [A]	188*12
Recommend discharging current [A]	157*12
Continuous Maximum Power [W]	0.5P (2507980.8W)
Discharge cut-off voltage [V]	1044.16
Temperature range for charging [°C]	1 to 55
Temperature range for discharging [°C]	-20 to 55
Temperature threshold for protection	56
Overcharge protected voltage supply by battery system	≥3.65V /Cell
Recommend charging method by manufacturer	After charging at a constant power of 0.5P until the maximum single-cell voltage reaches 3.55V, then charging at a constant power of 0.1P until the maximum single-cell voltage reaches 3.65V.
Dimension	6058mm×2438mm×2896mm
Weight	43.5t
Ingress Protection (IP)	Bat. Room: IP55 Electric room: IP55 Chiller controller module: IP55
Protective Class	I
Cooling type	Liquid cooling
Altitude	⩽3000m

Product	**Rechargeable Li-ion Battery System**
Type/model	Powtrix2.2-0.25P-Y104R04C12-A, Powtrix2.2-0.25P-Y104R04C12-B, Powtrix2.2-0.25P-Y104R04C12-C, Powtrix2.2-0.25P-Y104R04C12-D
Capacity [Ah]	314*12
Cell Quantity	4992
Battery structure	416S12P
Nominal voltage [V]	1331.2
Rated capacity [Wh]	5015961.6
Upper limit charging voltage [V]	1500
Maximum charging current [A]	94.2*12
Recommend charging current [A]	78.5*12
Maximum discharging current [A]	94.2*12
Recommend discharging current [A]	78.5*12
Continuous Maximum Power [W]	0.25P (1253990.4W)
Discharge cut-off voltage [V]	1044.16
Temperature range for charging [°C]	1 to 55
Temperature range for discharging [°C]	-20 to 55
Temperature threshold for protection	56
Overcharge protected voltage supply by battery system	≥3.65V /Cell
Recommend charging method by manufacturer	After charging at a constant power of 0.25P until the maximum single-cell voltage reaches 3.55V, then charging at a constant power of 0.1P until the maximum single-cell voltage reaches 3.65V.
Dimension	6058mm×2438mm×2896mm
Weight	43.5t
Ingress Protection (IP)	Bat. Room: IP55 Electric room: IP55 Chiller controller module: IP55
Protective Class	I
Cooling type	Liquid cooling
Altitude	⩽3000m

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
Chapter 6	**Fundamentals of Venting of Deflagrations**		--
6.1	**Basic Concepts.**		--
6.1.1	The deflagration index, K, shall be computed from the maximum rate of pressure rise attained by combustion in a closed vessel with volume, V, and shall be defined by the following equation [6.1.1]		P
6.1.2	For dusts, K_{St} and P_{max} shall be determined in approximately spherical calibrated test vessels of at least 20 L capacity per ASTM E1226, Standard Test Method for Explosibility of Dust Clouds.		N/A
6.1.2.1	It shall be permitted to determine K_{St} and P_{max} per ISO 6184-1, Explosion Protection Systems — Part 1: Determination of Explosion Indices of Combustible Dusts in Air.		N/A
6.1.2.2	The owner/user shall be permitted to test the dust with moisture content and particle size that deviates from the recommended conditions established by the method described in 6.1.2 or 6.1.2.1, provided a documented assessment acceptable to the authority having jurisdiction has been performed prior to using these K_{St} and P_{max} values to determine vent sizing.		N/A
6.1.2.3	For aluminum, hafnium, magnesium, tantalum, titanium, zirconium, and similar alloys or mixtures with adiabatic flame temperature higher than 3300°C, unless K_{St} and P_{max} are determined in nominal 1 m3 or larger calibrated test vessels, the K_{St} value shall be multiplied by a factor of 2 for application of the design methods.		N/A
6.1.3	For gases, P_{max} shall be determined in approximately spherical calibrated test vessels of at least 5 L (1.3 gal) capacity with initially quiescent mixture with low energy ignition source (i.e., less than 100 J).	See clause 4.	P
6.2	**Mixtures.**		--
6.2.1	Gas Mixtures.		P
6.2.1.1	Where the hazard consists of a flammable gas mixture, the vent size shall be based on the fundamental burning velocity of the mixture.	See clause 4.	P
6.2.1.2	Where the gas mixture composition is not certain, the vent size shall be based on the component having the highest fundamental burning velocity.		N/A
6.2.2	Dust Mixtures		N/A
6.2.2.1	Where the hazard consists of a dust mixture, the vent size shall be based on the K_{St} and P_{max} of the mixture.		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.2.2.2	Where the dust mixture composition is not certain, the vent size shall be based on the highest K_{St} of all components and the highest P_{max} of all components.		N/A
6.2.3	Hybrid Mixtures		N/A
6.2.3.1	For hybrid mixtures, the vent size shall be based on the equivalent mixture K_{St} as determined by test.		N/A
6.2.3.2	Where test data are not available for hybrid mixtures with gases that have combustion characteristics similar to those of propane (fundamental burning velocity ≤1.3 times that of propane) and St-1 and St-2 dusts, the design shall be permitted to be based upon P_{max} = 10 bar-g and K_{St} = 500 bar-m/s.		N/A
6.2.4	Foams of Combustible Liquids. Design of deflagration venting for foams of combustible liquids shall be based on tests performed on the specific foam.		N/A
6.3	**Enclosure Design and Support.**		--
6.3.1	Enclosure Design Pressure Selection Criteria.		P
6.3.1.1	P_{red} shall not exceed two-thirds of the ultimate strength for the vented enclosure, provided deformation of the equipment can be tolerated.	See clause 4.	P
6.3.1.2	Where deformation cannot be tolerated, P_{red} shall not exceed two-thirds of the yield strength for the vented enclosure.		N/A
6.3.1.3	For enclosures designed using the ASME Boiler and Pressure Vessel Code or similar codes, the maximum allowable working pressure, herein designated as P_{mawp}, shall be determined by calculation.		N/A
6.3.1.4	Ductile design considerations shall be used for materials subject to brittle failure, such as cast iron.		N/A
6.3.2	Venting shall be sufficient to prevent the maximum pressure that develops within the enclosure, P_{red}, from exceeding the enclosure strength, Pes, including the dynamic effect of the rate of pressure rise, as expressed by a dynamic load factor (DLF) [6.3.2]		N/A
6.3.2.1	In the absence of detailed structural response analysis, it shall be permitted to assume a worst-case value of DLF = 1.5 and design based on the weakest structural element of the enclosure.		N/A
6.3.2.2	It shall be permitted to modify the value of DLF based on a documented analysis of the vented explosion pressure profile and enclosure structural response.		N/A
6.3.3	All structural elements and supports shall be included in the design calculations.		P

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.3.3.1	The weakest structural element, as well as any equipment or other devices that can be supported by structural elements, shall be identified.		P
6.3.3.2	Where designing an enclosure to prevent catastrophic failure while still allowing permanent deformation, the normal dead and live loads shall not be relied on to provide restraint.		P
6.3.3.3	Structural members shall be designed to support the total load.		P
6.3.3.4	Doors, windows, ducts, or other openings in walls that are intended to be pressure resistant shall also be designed to withstand P_{red}.		P
6.3.4	Relieving Walls or Roof.		N/A
6.3.5	Enclosure Support Criteria.	Eliminated according to 6.3.5.4.1	P
6.3.5.4.1	The calculation of reaction forces on the enclosure shall be permitted to be eliminated when all of the following conditions are satisfied: (1) Vent panels are of the rupture diaphragm type. (2) Vent panels are located at opposing positions on the enclosure. (3) The P_{stat} of each vent panel is equal and less than or equal to 0.1 bar-g. (4) Vent panels are of equal area.		P
6.4	**Enclosure Length-to-Diameter Ratio and Vent Variables.**		--
6.4.1	The L/D of an elongated enclosure shall be determined based upon the general shape of the enclosure, the location of the vent, the shape of any hopper extensions, and the farthest distance from the vent at which the deflagration could be initiated.		N/A
6.4.2	For enclosures that can be vented at more than one point along the major axis, the vents shall be permitted to be distributed along the major axis and sized based on the length to diameter (L/D) between vents.		N/A
6.4.2.1	The maximum effective vent area at any point along the major axis shall be the enclosure cross section.		N/A
6.4.3	L/D of Elongated Enclosures.		N/A
6.4.3.1	The L/D of an elongated enclosure shall be determined based upon the general shape of the enclosure, the location of the vent, the shape of any hopper extensions, and the farthest distance from the vent at which the deflagration could be initiated.		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.4.3.2	The maximum flame length along which the flame can travel, H, shall be determined based on the maximum distance, taken along the central axis, from the farthest end of the enclosure to the opposite end of the vent.		N/A
6.4.3.2.1	When multiple vents are provided, a single value of H, and L/D, shall be permitted to be determined for the enclosure based on the farthest vent.		N/A
6.4.3.2.2	When multiple vents are located along the central axis, the value of H, and L/D, shall be permitted to be determined for each section using the maximum distance from the closest end of one vent to the opposite end of the next vent.		N/A
6.4.3.3	The effective volume of the enclosure, V_{eff}, shall be determined based on the volume of that part of the enclosure through which the flame can pass as it travels along the maximum flame length, H.		N/A
6.4.3.3.1	Partial volume (see Section 8.4) shall not be considered in the determination of effective volume per this section.		N/A
6.4.3.3.2	When multiple vents are provided, a single value of V_{eff} shall be permitted to be determined for the enclosure based upon the farthest vent.		N/A
6.4.3.3.3	When multiple vents are located along the central axis, V_{eff} shall be permitted to be determined for each section using the maximum distance from the closest end of one vent to the opposite end of the next vent.		N/A
6.4.3.3.4	When V_{eff} is less than the total volume of the enclosure, only those vents located within the effective volume shall be considered as providing venting for the event.		N/A
6.4.3.4	It shall be permitted to conservatively determine both H and V_{eff}, or H alone, but not V_{eff} alone, based on the total enclosure, irrespective of vent location.		N/A
6.4.3.5	The effective area, A_{eff}, shall be determined by dividing V_{eff} by H		N/A
6.4.3.6	The effective hydraulic diameter, D_{he}, for the enclosure shall be determined based on the general shape of the enclosure taken normal to the central axis [6.4.3.6]		N/A
6.4.3.6.1	Where the enclosure and any hopper extension are generally cylindrical, the perimeter, p, shall be permitted to be determined based on a circular cross section, given the following [6.4.3.6.1]		N/A
6.4.3.6.2	Where the enclosure and any hopper extension are generally rectangular or square, and the aspect ratio of the largest cross section is between 1 and 1.2, the perimeter shall be permitted to be determined based on a square cross section, given the following [6.4.3.6.2]		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.4.3.6.3	Where the enclosure and any hopper extension are generally rectangular, and the aspect ratio, R, of the largest cross section is greater than or equal to 1.2, the perimeter shall be permitted to be determined based on the aspect ratio of the largest cross section, given the following [6.4.3.6.3]		N/A
6.4.3.7	L/D for use in this standard shall be set equal to H/D_{he}.		N/A
6.4.4	The vent areas shall be permitted to be reduced from those specified in Chapters 7 and 8 if large-scale tests show that the resulting damage is acceptable to the user and the authority having jurisdiction.		N/A
6.4.5	The owner/user shall be permitted to install vents that are larger in area, are lower in density, or relieve at lower pressure than the minimum requirements determined from application of Chapter 7 or Chapter 8, as appropriate.		N/A
6.5	**Vent Closure Operation.**		--
6.5.1	The vent opening shall be free and clear.		N/A
6.5.2	Vent closure operation shall not be hindered by deposits of snow, ice, paint, corrosion, or debris, or by the buildup of deposits on their inside surfaces.		N/A
6.5.2.1	The materials that are used shall be chosen to minimize corrosion from process conditions within the enclosure and from ambient conditions on the nonprocess side.		N/A
6.5.2.2	Clear space shall be maintained on both sides of a vent to enable operation without restriction and without impeding a free flow through the vent.		N/A
6.5.2.3	To prevent snow and ice accumulation, where the potential exists, and to prevent entry of rainwater and debris, the vent or vent duct exit shall not be installed in the horizontal position, unless any of the alternative methods in 6.5.2.3.1 are followed.		N/A
6.5.2.3.1	Any of the following alternative methods of protection for horizontal vent or vent duct exits shall be permitted: (1) Fixed rain hats where Pred effects on vent area are included in accordance with Section 8.5 and restraint design includes maximum force from Pred applied over the area (2) Weather covers mounted at an angle sufficient to shed snow, with restraints designed and tested to prevent the cover from becoming a free projectile, where inertia effects of the additional weather cover mass and P_{stat} of the cover are included (3) Deicing provisions such as a heated vent closure		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.5.3	Restraining devices shall not impede the operation of the vent or vent closure device. (See Chapter 10.)		N/A
6.5.4	A vent closure shall release at its P_{stat} or within a pressure range specified by the vent closure manufacturer.		P
6.5.5	A vent closure shall reliably withstand pressure fluctuations that are below $P_{stat.}$		P
6.5.6	A vent closure shall withstand vibration or other mechanical forces to which it can be subjected.		P
6.5.7	P_{stat}, including the manufacturer's negative tolerance, shall be greater than the anticipated loading equivalent to the local design wind speed such that wind load will not cause the vent to open.		N/A
6.5.7.1	The area calculation shall be performed using the nominal value of P_{stat}.		P
6.5.8	P_{stat}, including the manufacturer's positive tolerance, shall be less than the intended P_{red}.		P
6.5.8.1	The area calculation shall be performed using the nominal value of P_{stat}.		P
6.5.9	Vent closures shall be maintained in accordance with Chapter 11		P
6.6	**Consequences of a Deflagration.**		--
6.6.1	The material discharged from an enclosure during the venting of a deflagration shall be directed outside to a safe location.		P
6.6.2	Property damage and injury to personnel due to material ejection during venting shall be minimized or avoided by locating vented equipment outside of buildings and away from normally occupied areas. (See Sections 7.6 and 8.9 for gases and dusts, respectively.)	See clause 4.	P
6.6.2.1	Deflagration vents shall not be located in positions closer to air intakes than the distances prescribed by the fireball length (see Sections 7.6 and 8.9).	See clause 4.	P
6.6.2.2	Deflagration vents shall be permitted to be located closer to buildings and normally occupied areas than the distances determined by Section 7.6 or Section 8.9, provided a documented risk assessment acceptable to the authority having jurisdiction has been performed.		N/A
6.6.2.3	Where a deflector is provided in accordance with 6.6.2.4 and 6.6.2.5, it shall be permitted to reduce the axial (front-centerline) hazard distance to 50 percent of the value calculated in 7.6.1 or 8.9.2. This method shall not be used to reduce the radial hazard distance as defined in 7.6.2 and 8.9.2.2.		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
6.6.2.4	A deflector design shall meet all of the following criteria: (1) The deflector for a rectangular vent shall be geometrically similar to the vent and sized with a linear scale factor of at least 1.75. For a round vent, the deflector shall be square shaped and at least 1.75 times the vent diameter. (2) The deflector shall be inclined 45 degrees to 60 degrees from the vent axis, as shown in Figure 6.6.2.4. (3) The centerline of the deflector shall be coincident with the vent axis. (4) The distance from the vent opening to the deflector on the vent axis shall be 1.5D, where D is the equivalent diameter of the vent. (5) The deflector plate shall be mounted so as to withstand the force exerted by the vented explosion, calculated as P_{red} times the deflector area. (6) The deflector location shall not interfere with the operation of hinged vent closures.		N/A
6.6.2.5	A deflector to limit flame length shall not be used as follows: (1) For enclosure volume greater than 20 m^3 (706 ft^3) (2) With a tethered or translating vent closure		N/A
6.6.3	Warning signs shall be posted to indicate the location of a vent.		P
6.7	**Effects of Vent Inertia.**		--
6.7.1	Counterweights and insulation added to panels shall be included in the total mass.		P
6.7.2	A vent closure shall have low mass to minimize inertia, thereby reducing opening time.		N/A
6.7.3	If the total mass of a closure divided by the area of the vent opening does not exceed the panel densities calculated by Equation 7.3.2 and Equation 8.3.2 (for gas and dust, respectively), all vent area correlations presented in this standard shall be permitted to be used without correction.		N/A
6.7.4	Hinged closures shall be permitted to be used, provided the following conditions are met: (1) There are no obstructions in the path of the closure that prevent it from opening. (2) Operation of the closure is not restrained by corrosion, sticky process materials, or paint.		P
6.8	**Fireball Dimensions.**		P

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

Clause	Requirement – Test	Result – Remark	Verdict
	Measures shall be taken to reduce the risk to personnel and equipment from the effects of fireball temperature and pressure.		
6.8.1	A documented risk assessment shall be permitted to be used to reduce the hazard distances calculated in 6.8.2, 6.8.3, and 6.8.7.		P
6.8.2	Gas Deflagration Fireball Dimensions.		P
	In the case of gas deflagration venting, L_F, the maximum axial length, width, and height of the fireball hazard zone distributed around the centerline of each vent discharge (see Figure 6.8.2) shall be expressed by Equation 6.8.2:		
6.8.3	Dust Deflagration Fireball Dimensions.		N/A
	In the case of dust deflagration venting, L_F, the maximum axial length, width, and height of the fireball hazard zone distributed around the centerline of each vent discharge (see Figure 6.8.2) shall be expressed by Equation 6.8.3		
6.8.4	Definition of Independent Vents.		N/A
	Independent vents shall meet the requirements in 6.8.4.		
6.8.4.1	For vents located at a single elevation and spaced around the circumference of a cylindrical vessel, the normal to the adjacent vents shall be separated by at least 60 degrees or 4 hydraulic diameters of the largest vent.		N/A
6.8.4.2	Vents located on different sides of a rectangular enclosure shall be considered independent.		N/A
6.8.4.3	For vents located along the axis of a cylindrical vessel or on the same side of a rectangular enclosure, the center of the vents shall be separated by at least four times the hydraulic diameter of the largest vent.		N/A
6.8.4.4	Vents that do not meet the requirements of 6.8.4 are allowed but shall not be counted separately to calculate n in 6.8.2 or 6.8.3.		N/A
6.8.5	Axial distance, calculated by Equation 6.8.3, shall be limited to 60 m [104].		N/A
6.8.6	Axial distance calculated by Equation 6.8.2 shall not be limited for gases.		N/A
6.8.7	Where venting is from a cubic vessel, the $P_{max,a}$ value shall be indicated approximately by Equation 6.8.7 [108]		N/A
6.8.8	For distances longer than $\alpha \times L_F$, the maximum external pressure, $P_{max,r}$, shall be indicated approximately by Equation 6.8.8		N/A

3. Deflagration venting design evaluation referring to Chapter 6 in NFPA68: 2023

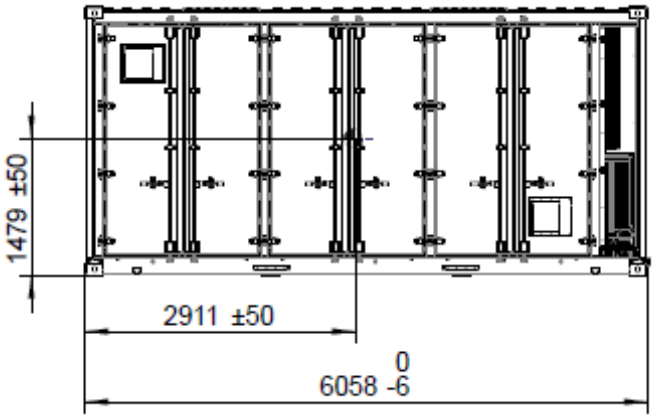
Clause	Requirement – Test	Result – Remark	Verdict
6.8.9	Equation 6.8.3, Equation 6.8.7, and Equation 6.8.8 shall be valid for the following conditions: (1) Enclosure volume: 0.3 m3 ⩽ V ⩽ 10,000 m3 (2) Reduced pressure: Pred ⩽ 1 bar-g (3) Static activation pressure: Pstat ⩽ 0.1 bar-g (4) Deflagration index: KSt⩽ 300 bar-m/s for Equation 6.8.3, KSt ⩽ 200 bar-m/s for Equation 6.8.7 and Equation 6.8.8 (5) Pmax ⩽ 9 bar-g		N/A
6.9	**Effects of Vent Discharge Ducts.**		N/A
6.10	**Venting with Flame Arresting and Particulate Retention.**		N/A

4. CFD analysis of the ventilation system

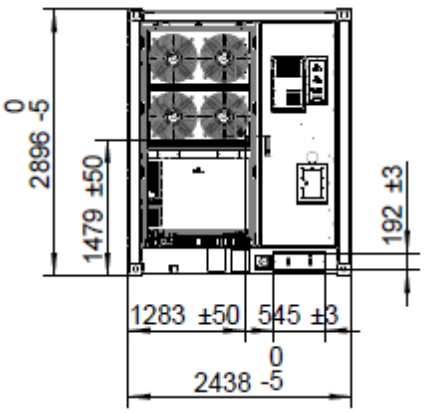
4.1. Basic Information of the system

The interior and exterior drawings provided by REPT were verified between TUV and REPT before running any calculation.

Figure 4-1 shows an external view of the container and the dimensions from a front and left view. The cube measures 6.058*2.438*2.896 m including one battery cabinet, liquid coolant unit, BCP and Auxiliary power supply unit. The cabinet contains 48 modules in rack and 104 cells in each module. The overall internal volume of battery cabinet is approximately 31.20 m^3. Subtracting the space filled by racks, modules, and the support structure, the actual open internal volume (efficient volume) is 9.78 m^3.



Front view



Right view

Figure 4-1: Geometry of REPT energy storage system

4.2. UL 9540A Test Summary

The gas data used in the NFPA 69 test is based on cell level, module level and unit level UL 9540A test reports.

4.2.1.Gas composition

Gas composition was provided in the test data of cell level UL 9540A report. Table 4-1 lists the release of gas composition by volume and Table 4-2 lists the flammability characteristics.

Table 4-1: Gas composition by volume

Gas component	Concentration % (v/v)
CH_4	5.480
C_2H_2	0.168
C_2H_4	2.838
C_2H_6	0.804
C_3H_4	0.011
C_3H_6	0.428
C_3H_8	0.324
C_4H_6	0.323
C_4H_8	0.279
C_4H_{10}	0.263
H_2	54.373
CO	10.095
CO_2	24.614
Total	100

Table 4-2: Cell Flammability Properties

Average cell surface temperature at gas venting :	127.9 °C
Average cell surface temperature at thermal runaway :	179.8 °C
Gas Volume:	188 L
Lower flammability limit (LFL), % volume in air at the ambient temperature:	7.7 %
Lower flammability limits (LFL), % volume in air at the venting temperature:	7.1 %
Burning velocity (Su) :	0.688 m/s
Maximum pressure (Pmax) :	0.798 MPa

The offgas temperature is 127.9°C, by the cell-level UL 9540A test report.

4.2.2. Gas cloud and ignition location

(1) Gas cloud:

Based on the simulation results of NFPA 69, the maximum average gas concentration ratio is 0.74% (See figure 4-2). Taking the net volume of the container into account, the maximum gas volume during the entire emergency ventilation process is calculated to be 0.072 m³. Due to the sparse distribution of combustible gases and the gas concentration being below the LFL during the emergency ventilation process, ignition is comparatively difficult.

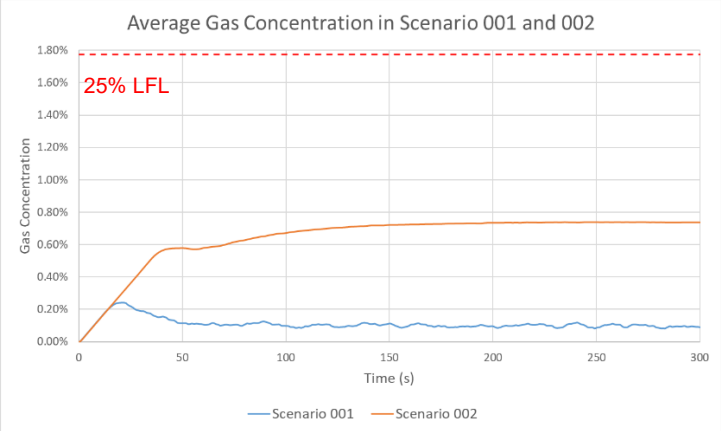


Figure 4-2: Average gas concentration plot

The homogeneous stoichiometric cloud considered in this analysis serves as a conservative estimate of the overpressures generated, as the combustion characteristics of such a cloud would yield the highest explosion overpressures with increased flame speeds.

Based on the gas content reported in the UL 9540A cell-level report, the volume of the stoichiometric concentration gas cloud is calculated to be 0.31 m³, assuming the gas cloud is uniformly distributed around the ignition point.

(2) Ignition location:

The ignition location was selected to be a single point in front of the modules and towards the top. The fire and pressure wave travel further when gas cloud ignited near corner, so top corner ignition was chosen. Since all the wiring and openings to the modules are on the front, it is assumed that the ignition sources are most likely to occur in front of the modules. However, we set another ignition point in the top center of the gas cloud for comparation where the local concentration is the highest in the cabinet (See figure 4-3).

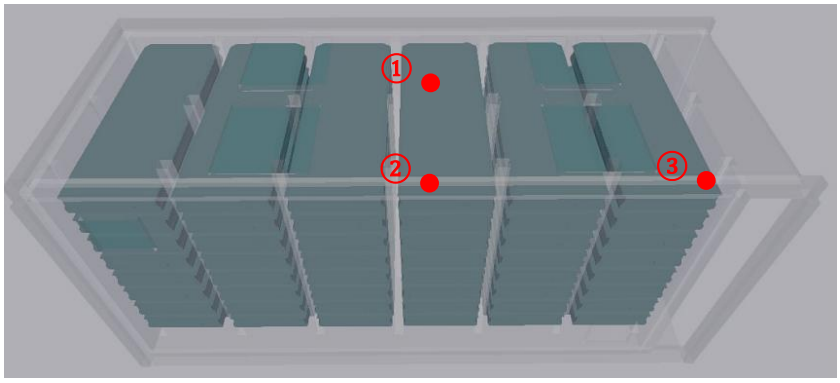


Figure 4-3: Ignition positions

4.3. Deflagration venting system

Deflagration venting is considered to occur through the explosion relief panels placed on the roof of the enclosure. Details of the considered explosion relief vent design are in Table 4-3. These vent characteristics are the same for all the vents considered in this work.

Table 4-3: Deflagration panel information

Manufacturer name	Type number	Dimension	Trigger pressure
IEP Technologies	KE 750X840	750*840mm	0.1 bar
Dalian DUTA Safety Technology Co., Ltd.	VPF840*750	750*840mm	0.1bar

Four deflagration panels are installed on the container, 0.750*0.840 m effective area for each panel and act at 0.1 bar pressure difference (See figure 4-4).

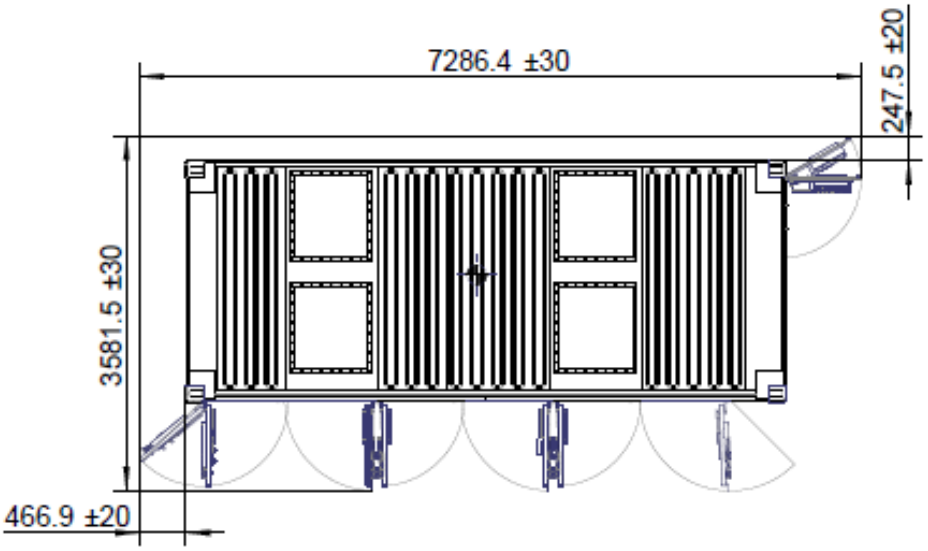


Figure 4-4: Deflagration venting layout

5. CFD MODEL APPROACH FLACS ANALYSIS

5.1. Design Scenario

5.1.1. Pre-Processor

CASD is an acronym for Computer Aided Scenario Design. The preprocessor CASD for the CFD simulator FLACS-CFD is used to prepare the input data, or job data, that defines a FLACS-CFD simulation. CASD provides a visual depiction of the developed model geometry and enables accurate meshing in the model to avoid any numerical instabilities causing effects from the FLACS input file.

5.1.2. FLACS Solver

Computational fluid dynamics (CFD) is a branch of fluid mechanics that uses numerical methods and algorithms to solve and analyze problems that involve fluid flow, with or without chemical reactions. Current use of CFD covers a broad range of applications, from fundamental theoretical studies involving models primarily derived from first principles to practical engineering calculations utilizing phenomenological or empirical correlations.

Many of the hazards encountered in society, and especially in the process industries, involve accident scenarios where fluid flow in complex, large-scale, three-dimensional (3D) geometries play a key role.

FLACS-CFD is a specialized CFD toolbox developed especially to address process safety applications such as:

- Dispersion of flammable or toxic gas
- Gas and dust explosions
- Propagation of blast and shock waves
- Pool and jet fires.

The development of FLACS-CFD started in 1980 at the Department of Science and Technology at Christian Michelsen Institute (CMI). CMI established Gexcon (Global Explosion Consultants) as a consultancy activity under the Process Safety Group in 1987. In 1992, the Science and Technology department at CMI became Christian Michelsen Research (CMR), and CMR established Gexcon as a private limited company in 1998.

5.1.3. Post Processing & Visual Representations

Flowvis, the postprocessor for the CFD-code FLACS-CFD, is a program for visualizing results from simulations of gas explosions, gas dispersion and multiphase flow.

5.2. Geometry and Grid Size

In this analysis, the BESS enclosure is modeled based on the 3D model for the battery module storage area. (See Figure 5-1)

Based on the overall dimensions of the container, and to ensure the accuracy of the simulation, the core grid size is set to 0.04 m. The grid resolution has been specifically determined for such BESS container ventilation analyses, ensuring it is appropriate to capture the necessary fluid dynamics and details of the flow conditions within the container,

while providing sufficient resolution across the modeling surfaces. The grid is extended 5 m in both positive and negative directions along the x-axis and y-axis. Along the z-axis, it is extended to 10 m in the positive direction, while no extension is applied in the negative direction.

Standard temperature and pressure conditions of 25°C and 1.03 bar respectively were assumed throughout this analysis unless otherwise noted. This analysis does not consider site-specific conditions and is mainly focused on the BESS enclosure, it's general configuration and operation.

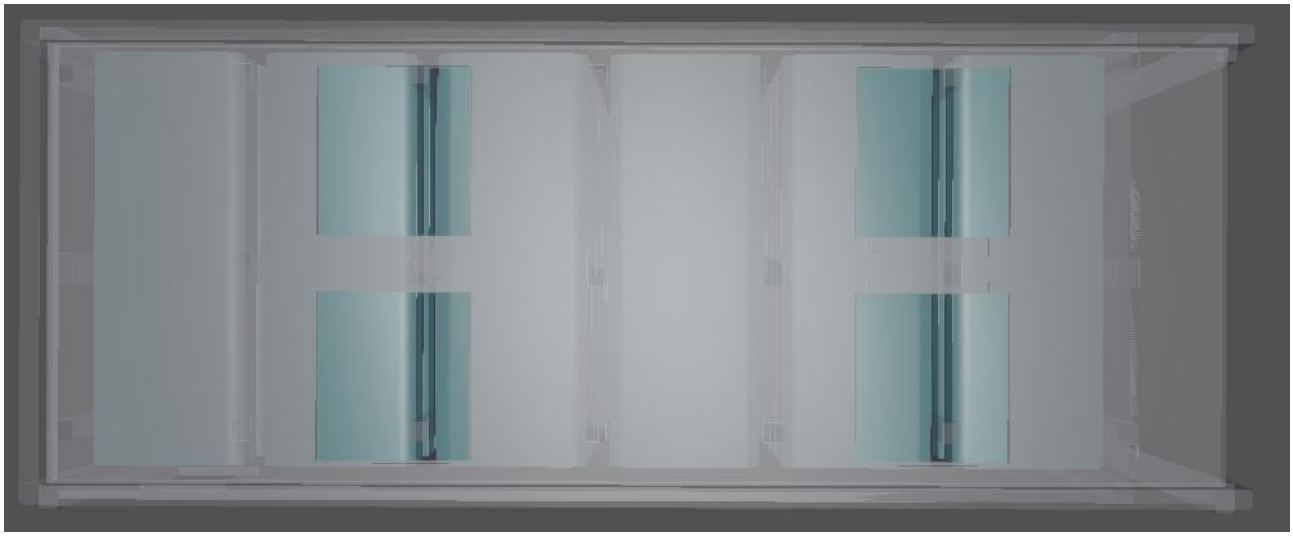


Figure 5-1: REPT Enclosure BESS FLACS Model (top view)

6. Assumptions and limitation

6.1. Assumptions

The CFD analysis performed in this study relies on several assumptions that are listed below. These assumptions are considered to add some conservatism to the modeling methodology and adhere to the FLACS validation range. Modeling assumptions include:

The BESS enclosure is considered a rigid structure

The fuel cloud is of homogeneous stoichiometric concentration

The battery racks are treated as rigid obstacles

The ventilation system inside the BESS enclosure is assumed to be switched off during the event

6.2. Limitation

The explosion event is considered in isolation and does not include domino effects such as fire and the potential initiation of thermal runaway in other adjacent battery racks.

7. Results and Conclusions

Results from the partial volume deflagration analysis for the different ignition locations considered (see Section 4.2.3) are presented in this section. The evaluation focuses on the generated overpressures inside the

BESS container from the deflagration event, given that the maximum pressure limit was bound to 0.50 bar-g (as per Client input).

7.1. Simulation Results

A total of three gas explosion scenarios were run representing progressively worse-case scenarios based on the assumptions made in the previous sections. The modelling covered the failure of 3 cells with different ignition positions.

Table 7-1 presents a summary of the scenarios and the results from the simulation. The maximum over pressure inside enclosure is presented according to NFPA 68.

Table 7-1: Maximum wall pressure inside enclosure

Scenario	Ignition Position	Maximum pressure (bar-g)
001	1	0.269
002	2	0.155
003	3	0.162

(1) Pressure curve

Figure 7-2 to 7-4 shows the wall pressure for each scenario.

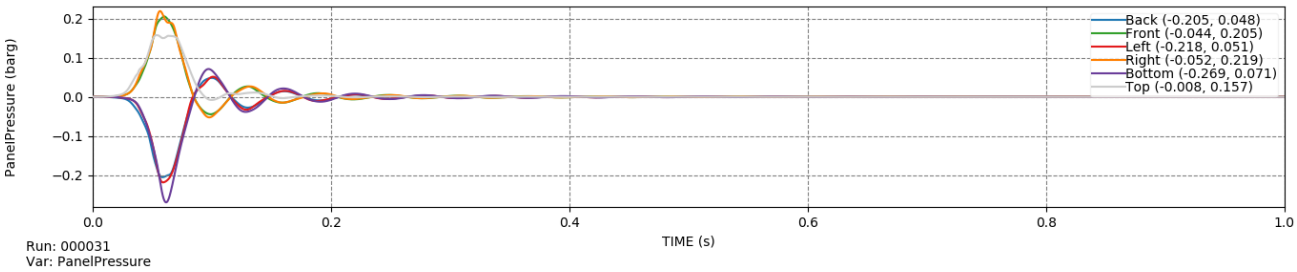


Figure 7-2: Top center ignition pressure curve.

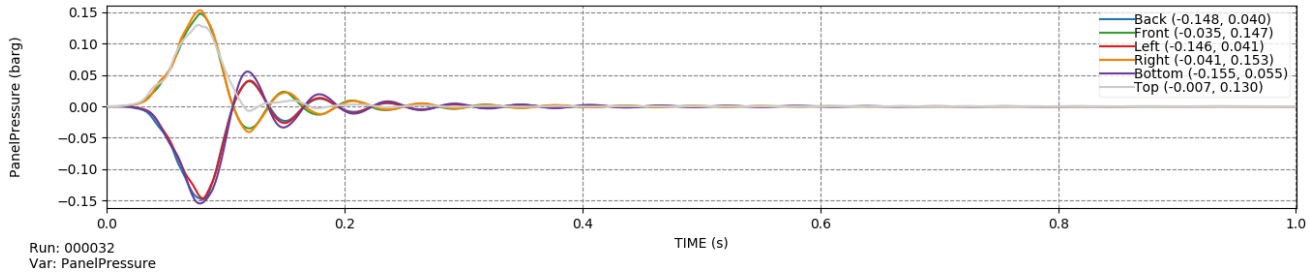


Figure 7-3: Top front center ignition pressure curve.

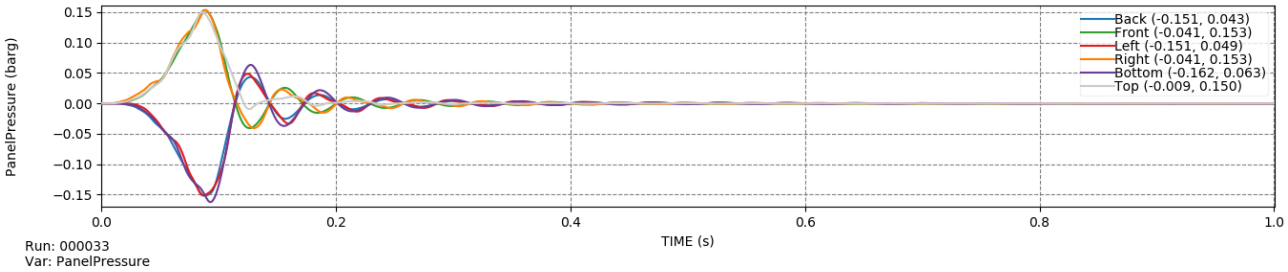


Figure 7-4: Top front corner ignition pressure curve.

Figure 7-5 and 7-7 shows the maximum pressure outside cabinet.

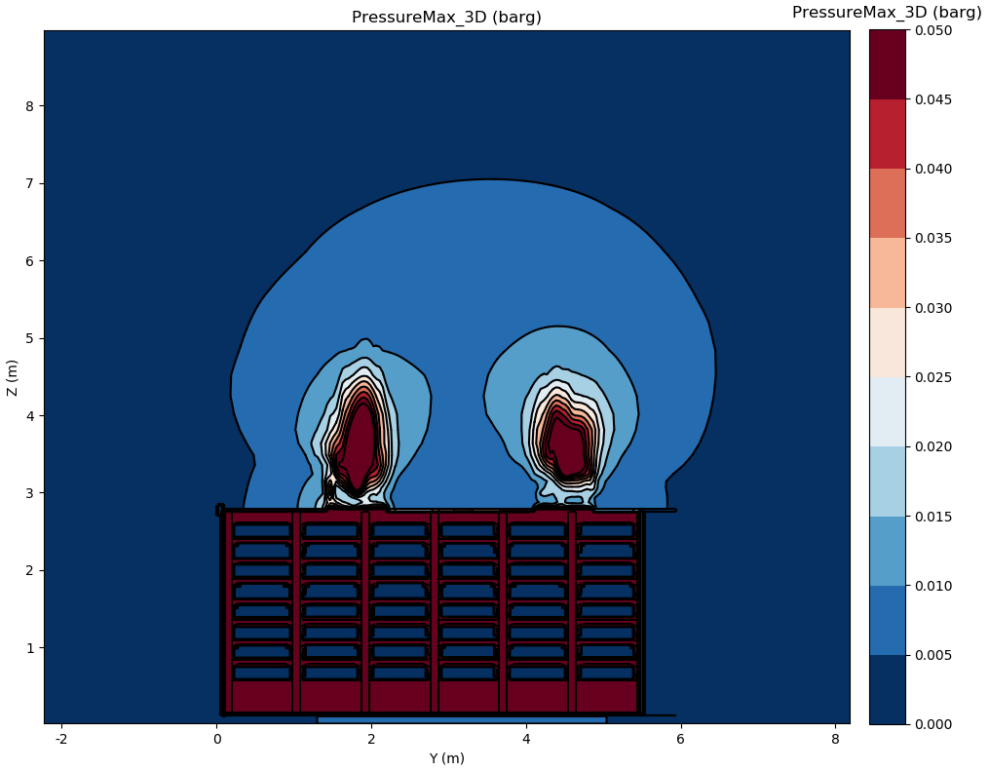


Figure 7-5: Top center ignition pressure map (outside).

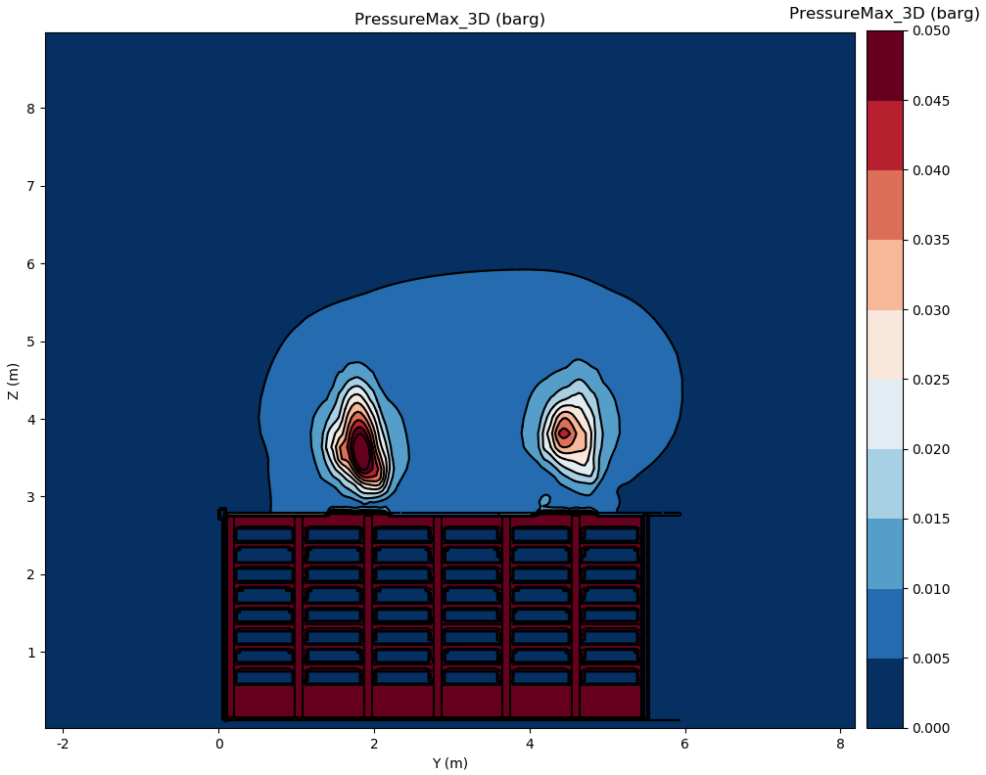


Figure 7-6: Top front center ignition pressure map (outside).

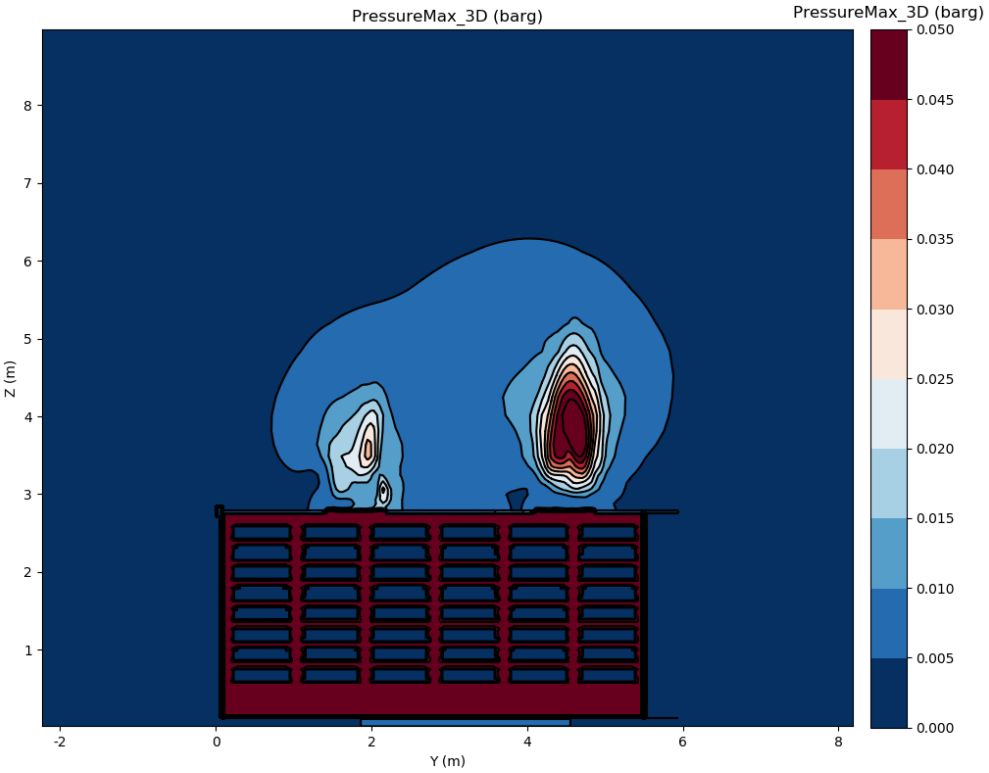


Figure 7-7: Top front corner ignition pressure map (outside).

(3) Temperature map

Figure 7-8 and 7-10 show the temperature map.

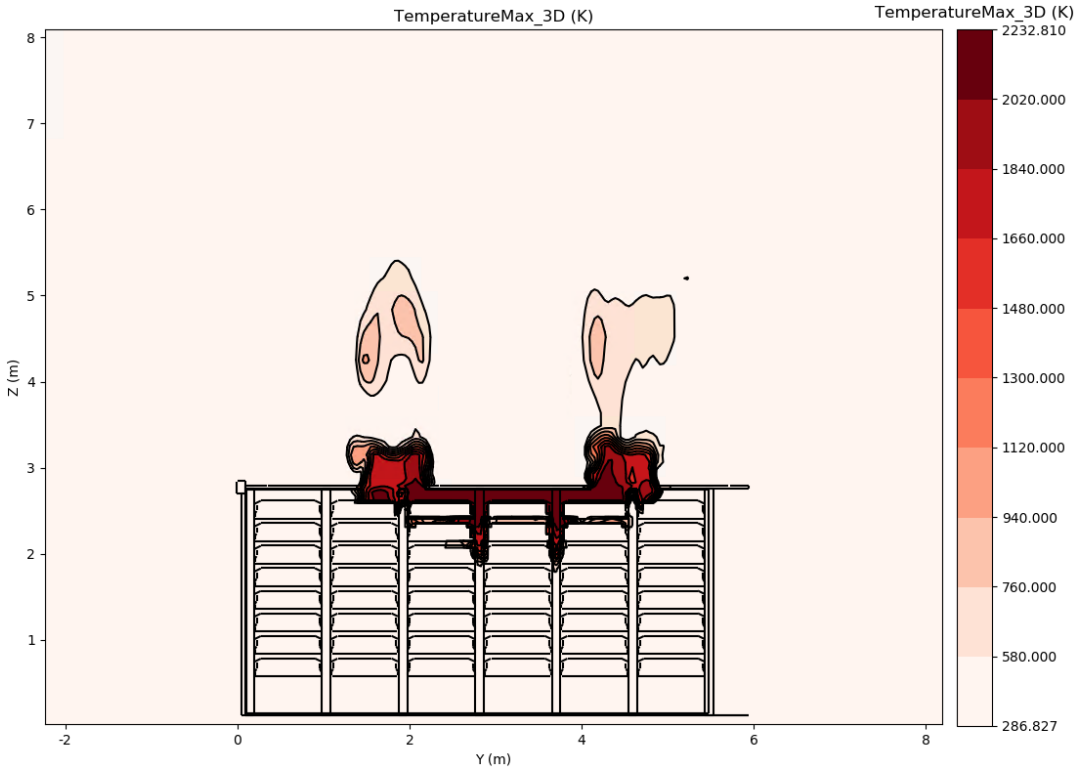


Figure 7-8: Top center ignition temperature map (outside).

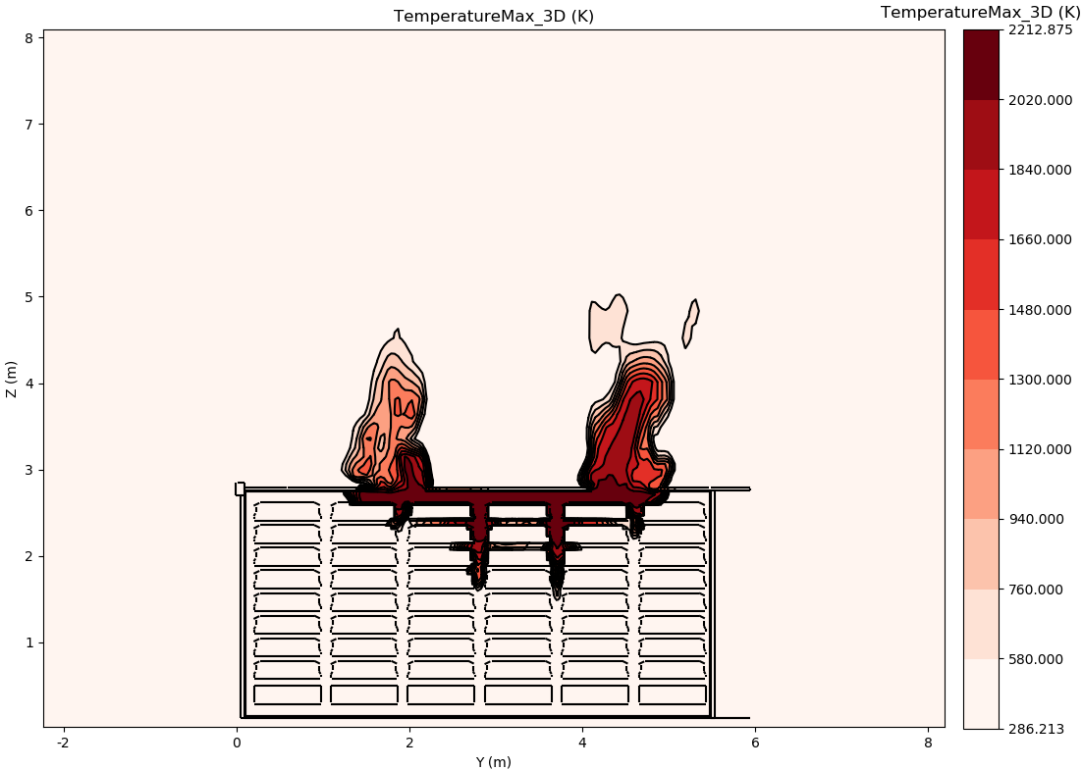


Figure 7-9: Top front center ignition temperature map (outside).

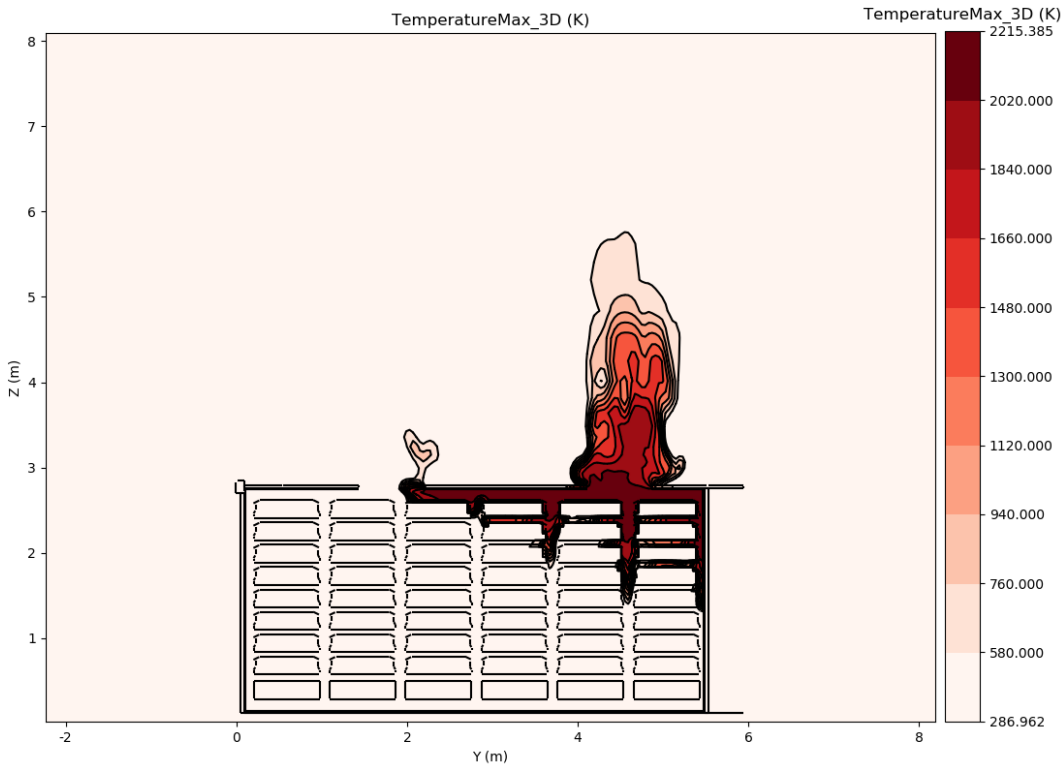


Figure 7-10: Top front corner ignition temperature map (outside).

7.2. Conclusions

This CFD model has shown that the predicted maximum average pressure on the wall is 0.269 bar-g. The enclosure mechanical analyze report is provided by REPT, and REPT is responsible for the enclosure mechanical strength. The enclosure mechanical analysis report indicates that the enclosure could maintain at least 0.50 bar-g of pressure. According to section 6.3.1.1 P_{red} shall not exceed two-thirds of the ultimate strength for the vented enclosure, provided deformation of the equipment can be tolerated. Therefore, P_{red} shall not exceed 0.333 bar-g, which confirms that this design meets the requirements of NFPA 68.

8. References

(1) NFPA68:2023 Standard on Explosion Protection by Deflagration Venting

(2) Cell Level UL9540A Report: CN234568 001

(3) Module Level UL9540A Report: CN241BQ6 002

(4) Unit Level UL9540A Report: CN240QMQ 002

(5) NFPA 69 Report: CN26B0L6 001

(6) Explosion Simulation Report from REPT

End of Report